Diffusion Mechanism and Knowledge Distillation Object Detection in Multimodal Remote Sensing Imagery

Chenke Yue, Yin Zhang, Junhua Yan, Zhaolong Luo, Yong Liu, and Pengyu Guo

***Abstract*— Multimodal remote sensing images provide complementary information, enhancing the effectiveness of object detection tasks in open-world scenarios. To address the imbalance of information richness between modalities in multimodal object detection, we propose a simple yet effective multisource image object detection method (DKDNet). Our contributions are twofold: 1) we introduce diffusion deformation convolution (DDConv), which combines deformation convolution with adaptive long-range receptive fields to further enhance the ability to perceive object pose variations and capture distant information. 2) We propose the bidirectional feature distillation and information complementary fusion network (BDFusion), where different modalities exchange information through a knowledge distillation strategy, explicitly enhancing the information interaction between modalities. Finally, we adaptively build spatial domain complementarity between different modalities via self-correction, revealing implicit correlations. Experimental results on the publicly available vehicle detection in aerial imagery (VEDAI) dataset and the optical and synthetic aperture radar (SAR) ship detection dataset (OSSDD), collected in the Suez Canal region, demonstrate that our proposed method achieves superior performance with acceptable inference time, making it suitable for various real-world scenarios.**

***Index Terms*— Diffusion deformation, feature distillation, implicit correlations, mutually inject, optical and synthetic aperture radar (SAR), spatial domain complementarity.**

I. Introduction

OBJECT detection is a crucial task in perceiving and applying remote-sensing images. It plays a significant role in various application fields such as defense, urban planning, disaster assessment, and navigation by identifying and locating objects in remote sensing images [1], [2], [3]. However, relying solely on single-modal data for object detection [4], [5] can be affected by the imaging mechanisms of remote sensing images. For example, optical images are susceptible to illumination and atmospheric conditions, while synthetic aperture radar (SAR) images are affected by speckle noise due to their imaging mechanisms. As shown in Fig. 1(a), optical images are affected by weather and illumination conditions, causing objects to appear with shadows or be obscured by clouds and fog, making it difficult to distinguish between the background and the target as well as the geometric contours of the target itself. In contrast, SAR images, which rely on objects' radiation characteristics, can highlight objects from the background. As shown in Fig. 1(b), when lighting conditions are favorable, optical images can provide the target's clear color, shape, and geometric features. Still, these texture features are challenging to obtain in SAR images.

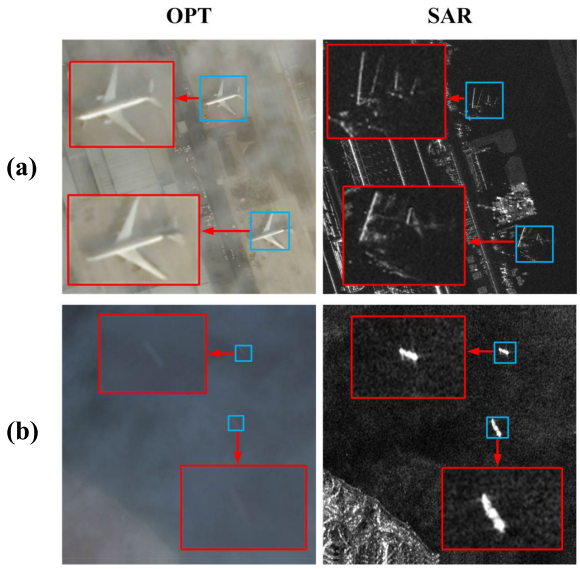


Fig. 1. Paired examples of optical and SAR images. (a) Images of Shanghai airport. (b) Section of a fog-covered scene over the Suez Canal.

Remote sensing big data enables the acquisition of images through multiple observation methods in the same target area, facilitating the exchange of information between different modalities. This complementary advantage greatly enhances the accuracy of target recognition and the precision of localization, making the effective fusion of multimodal features crucial for remote sensing image target detection tasks. Initially,

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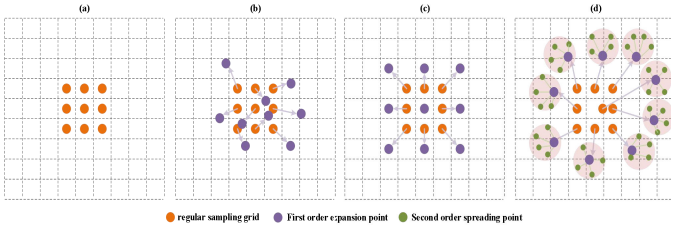


Fig. 2. Illustration of sampling positions in 3 × 3 standard convolution, dilated convolution, DConv, and DDConv. (a) Regular sampling grid of standard convolution (orange dots). (b) Deformed sampling positions of deformation convolution (purple dots). (c) Sampling positions of dilated convolution (purple dots). (d) Sampling positions of DDConv, including first-order points (purple) and diffused second-order points (green), indicating the ability of DDConv to capture various transformations of different aspect ratios and shapes at surrounding and long-range distances.

traditional methods [6], [7] relied on manually extracting target features. With deep learning demonstrating its powerful feature extraction capabilities, multimodal target detection methods based on deep learning have begun to dominate the mainstream. The initial approach of fusing multimodal images for deep learning detection tasks [8], [9] involved overlaying two modalities at the pixel level and feeding them into a neural network for detection and recognition. Subsequently some methods [10], [11], [12] treat different modal features equally, reconstructing spatial and channel domains through gating and attention mechanisms before fusing multimodal features.

However, given the limited receptive field of convolutional operations, convolution-based methods inherently capture only local features, thus lacking the ability to model global information. You et al. [13] addressed this limitation by adopting TransFormer [14] as an alternative to convolutional operations. Despite mitigating the issue of insufficient global information, the substantial stacking of TransFormer blocks may introduce computational overhead and redundant information. Some methodologies [15], [16] have proposed lightweight TransFormer structures to minimize computational overhead and redundancy. However, due to the necessity of perceiving global information, challenges such as substantial computational demands and redundancy persist.

Inspired by the deformable convolutional networks (DCNs) [17], we propose an enhancement based on deformable convolution (DConv), denoted as diffusion deformation convolution (DDConv). As depicted in Fig. 2, the designated convolutional layer learns the offsets of each sampling grid from the feature map itself. These computed offsets are then added to the sampling grid positions of the standard convolution to obtain first-order sampling points. Building upon these first-order sampling points, diffusion sampling is performed on the surrounding areas to obtain second-order sampling points. Subsequently, the similarity between the two sets of sampling points and the convolutional center is computed to capture long-range dependencies.

After extracting feature information from different modalities, how to fuse and detect the complementary information from these modalities is crucial for multimodal object detection tasks. Most fusion methods focus on linearly concatenating the features from different modalities, followed by modeling in both spatial and channel dimensions [18], [19]. However, these methods overlook the significant differences in the richness of information carried by each modality, which results in a low utilization rate of multimodal information.

We propose a bidirectional feature distillation and information complementary fusion network (BDFusion) for fusing multimodal features to explicitly integrate the complementary nature of different modality features. The proposed framework comprises two primary components: bidirectional knowledge distillation interaction and adaptive spatial domain reconstruction module. Initially, distinct branches for knowledge injection are established across different modalities to facilitate dual-branch fusion, enabling the integration of current modality information into others. This fusion strategy distinctly prioritizes primary and secondary components, allowing for extracting missing information from other modalities and visibly enhancing intermodality information utilization. Recognizing the potential limitations of the knowledge injection branch in effectively representing the current modality's information, we adopt a knowledge distillation strategy [20] to guide shallow knowledge injection branches in effectively capturing modality features using pretrained deep neural networks. Upon receiving complementary information from other modalities, the adaptive spatial domain reconstruction Module generates spatial weight coefficients for different modality features, selectively fusing them in the spatial domain using a hard-threshold gate mechanism. In summary, our contributions are the following.

1) We introduce a DDConv to extract long-range dependencies and geometric transformation features, capturing global contextual information.
2) We present a BDFusion to integrate multimodal features. It primarily consists of two components: 1) bidirectional knowledge distillation interaction and 2) adaptive spatial domain reconstruction module.
3) The proposed feature distillation scheme can guide the learning ability of the modal injection branch through a pretrained backbone, improving the utilization rate of intermodal correlations.
4) Our method demonstrates state-of-the-art results in experiments conducted on real-world RGB + IR multimodal vehicle detection in aerial imagery (VEDAI) [21] vehicle dataset and RGB + SAR multimodal optical and SAR ship detection dataset (OSSDD) ship dataset.

This article is organized as follows: Section II presents related works. Section III provides a detailed description of the proposed methods. Section IV outlines the experimental configurations and details, presenting the experiments' results and a thorough discussion. Finally, Section V concludes the article and discusses future research directions.

II. Related Work

A. Object Detection Based on RGB and IR Images

In real-world scenarios under complex conditions, single-modal data are often affected by their own imaging mechanisms, resulting in problems such as occlusion or unclear

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imaging of object targets. Given the complementary advantages of multimodal data, many researchers have proposed multimodal object detection methods to overcome the challenges of single-modal data. Qingyun and Zhaokui [22], considering the complementary information of RGB and IR images to enhance detection capability, proposed an adaptive feature enhancement and reconstruction method based on common feature attention maps and differential feature attention maps of the two modal images, which are fused with the source images for detection. The one-stage detector YOLO [23] due to its outstanding performance in natural images, has been adopted as a benchmark detection model in many methods, with fusion modules introduced in intermediate output layers for improvement [24]. SuperYOLO [25], based on YOLOv5 as the detection framework, obtains additional data information through a compact multimodal fusion approach.

Numerous methods have been proposed for pedestrian and salient object detection using IR data as supplementary information to RGB datasets. For example, Kim et al. [26] addressed the issue of poor fusion effects due to different imaging perspectives from various data sources. Deevi et al. [27] constructed a lightweight multimodal detection architecture using a pretrained single-modal detection backbone network combined with small fusion modules, avoiding retraining the entire network. In multimodal salient object detection tasks, many methods [28] focus on designing multilevel encoding for different modal data, followed by a fusion of encoded features and decoding for detection. Unlike dual-branch encoding designs, Huo et al. [29] adopted a three-branch encoding to comprehensively explore a single modality, merging complementary multiscale information of RGB and infrared images using separate branches.

B. Multimodal Data Fusion Based on Optical and SAR

Using only multispectral data makes differentiating features in regions with spectral similarity challenging. Typically, the fusion of SAR or depth maps with optical data addresses this issue. However, when using deep learning to extract and fuse features from both, the heterogeneity and nonlinear radiometric differences between optical and SAR images significantly reduce the effectiveness of the fused features. Moreover, overly focusing on the fusion of complementary information can inevitably lead to a loss of structural consistency between the two modalities. Therefore, the key to optical and SAR feature fusion lies in balancing the extraction of their complementary and consistent features.

Due to the greater difficulty in acquiring SAR data compared to optical data, some methods enhance detection accuracy by generating SAR [30] or polarimetric SAR [31] data. It is important to note that inaccurate registration between the two modal images, which lead to misalignment in the target regions of the modalities, can significantly degrade detection accuracy [32]. Similarly, the fusion of SAR and optical images presents a considerable challenge due to the significant heterogeneity between the two modalities and the presence of speckle noise in SAR images [33]. Ye et al. [34] recorded features into structural and detail features and refused them separately. Particularly, they utilized Gabor wavelet feature descriptors to strip off noise from SAR images, selectively retaining their intrinsic radiometric feature information. Li et al. [35] adjusted optical and SAR images to the same feature space through domain transformation, making the feature distributions similar and facilitating subsequent fusion. Shermeyer et al. [36] established a multimodal optical and SAR data dataset and fused different modality features. Subsequently, Adriano et al. [37] analyzed various building damage assessment scenarios and developed a multimodal damage mapping model. Li et al. [38] proposed a progressive fusion learning framework, utilizing phase information from optical and SAR images as explicit features of high correlation between the two modalities, serving as the information medium for fusion between modalities.

Some methods have also introduced contrastive learning [39], [40] into cross-modal feature learning between optical and SAR data. BCLNet [41] employs contrastive learning to learn the feature space of both modalities across the entire dataset and utilizes boundary point hard negative mining to enhance the feature embedding. BIDNet [42] employs a teacher–student framework with distinct explicit and implicit distillation modules to capture clear boundary information from optical images. Lee et al. [43] utilized the rich contextual information extracted from optical features to assist unlabeled SAR images in weakly supervised learning, thereby enhancing the extraction of SAR features. Similarly, in scenarios with limited labeled samples, Protopapadakis et al. [44] applied weakly supervised learning to generate soft labels through a series of soft estimates, enabling the training of deep models on many unlabeled samples.

In multimodal object detection tasks, whether it pertains to the fusion of RGB and IR images or optical and SAR images, leveraging the distinctive imaging characteristics between different modal images is paramount for effective feature fusion. Within the feature fusion phase, we introduce the cross-modal BDFusion. By incorporating injected information from the alternate modality and employing knowledge distillation strategies to enhance the learning capability of modality features.

III. Methodology

We present DKDNet, a novel approach designed for multimodal object detection aimed at refining the precision of object detection within real-world settings via feature-level fusion. Fig. 3 illustrates the YOLOv8 network we adopt as the baseline model, consisting primarily of three parts: Backbone, Neck, and Head. In DKDNet, we first employ a dual-path backbone network to separately extract multilayer effective features from different modal images, enhancing the utilization efficiency of heterogeneous images. We divide the training phase into two parts: in the first phase, training is performed without a knowledge distillation strategy. In the second phase, the backbone network trained in the first phase serves as the teacher model, and bidirectional injection branches are applied for distillation.

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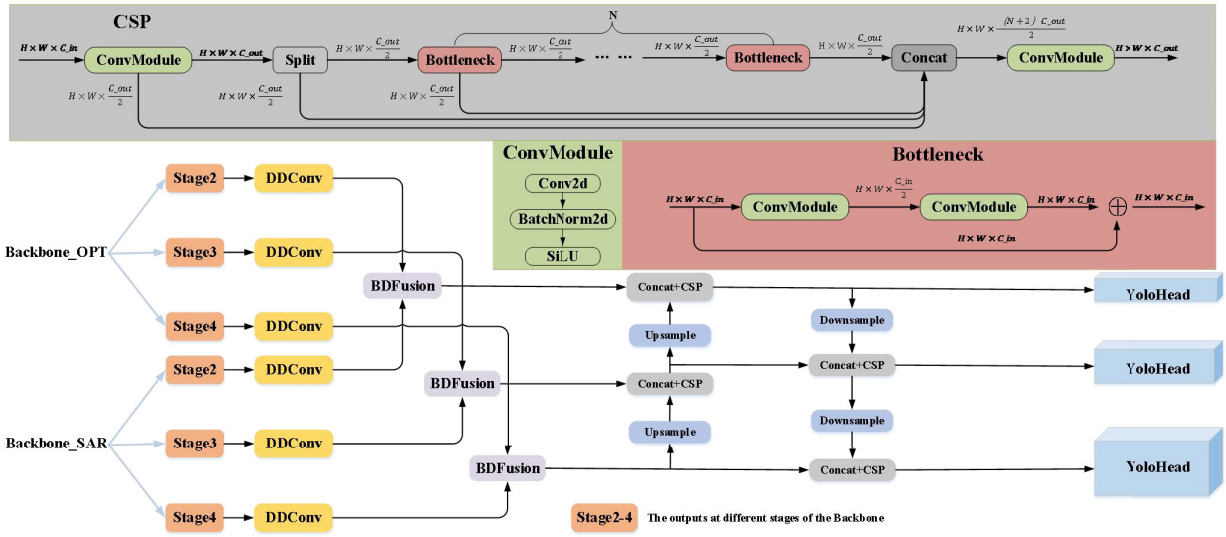


Fig. 3. Overall architecture of DKDNet. Our DKDNet adopts a dual-branch backbone network to extract deep and shallow features from two modalities. Subsequently, each layer's features are enhanced through DDConv for feature extraction, followed by multimodal feature fusion via the BDFusion.

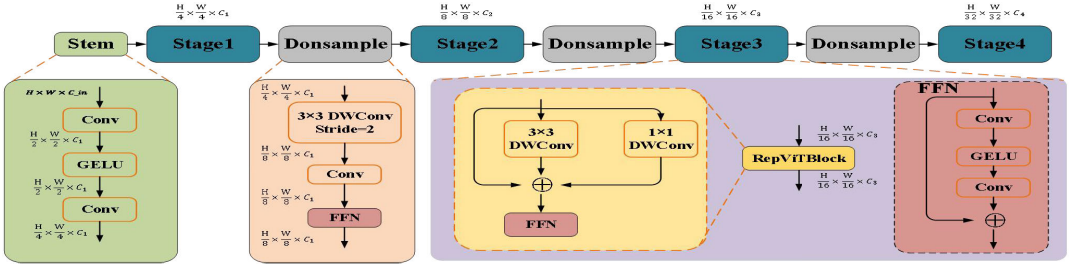


Fig. 4. Backbone network structure of RepViT [45]. Intermediate outputs from the three modules in stages 2–4 are selected as the three effective features extracted for the modalities. Stages 2–4 are stacks of 3, 14, and 2 RepViTBlocks.

A. Overall Framework

As shown in Fig. 3, in the feature extraction part, both deep and shallow features from the two modal images are simultaneously extracted by passing them through the backbone using a dual-input stream method. The backbone utilizes the RepViT [45] method, which combines a lightweight convolutional neural network (CNN) and lightweight ViTs [46] architecture. Its reparameterization enhances computational efficiency, enabling efficient feature extraction and strong transfer learning capability. As shown in Fig. 4, RepViT [45] mainly consists of four stages: Stem, stages 1–4, Pooling, and FC. For the purpose of reducing the network's computational complexity, we select the first two stages, discard the SE module, and use only the RepViTBlock to form the stage phase, thereby constructing the backbone. Its formal representation is as follows:

$$F_a^i,\ F_b^i = \Phi_{\text{backbone}}(I_a),\ \Phi_{\text{backbone}}(I_b)$$

where $F_a^i, F_b^i \in \mathbb{R}^{C_i \times H \times W}$ are two different modality feature maps extracted from the middle output of the backbone at stage i ($i \in \{1, 2, 3, 4\}$). Here, $C \in \{48, 96, 192, 384\}$ represents the number of feature channels. To reduce the computational burden of the model and accelerate inference speed, we select stages 2–4 for modal feature extraction, starting from the downsampling to one-eighth of the original image size. H and W represent the height and width of the feature map, respectively. Each of the stages 2–4 modules consists of 3, 14, and 2 RepViTBlocks stacked together. Subsequently, the feature maps obtained from each modality are passed through DConv layers to extract diverse target dispersion area information, building dependencies between long-range targets. The detailed design will be introduced in Section III-B

$$S_x^i = \text{DDconv}\left(F_x^i\right), \quad x \in \{a, b\}.$$

Next, in multimodal object detection tasks, it is necessary to merge the feature maps of two different modalities, S_a^i and S_b^i. We adopt an intermediate feature fusion strategy, proven in extensive prior research to balance speed and accuracy well. Using bidirectional modalities, the details of the fusion method will be elaborated in Section III-C

$$S_{a+b}^i = \text{BDFusion}\left(S_a^i, S_b^i\right).$$

B. DDConv

Although popular transformer-based methods and their variants have been proven effective in capturing global contextual information and establishing long-range dependencies, they overlook the issue of feature locality. Therefore, considering the diversity of target sizes and the main challenges in current

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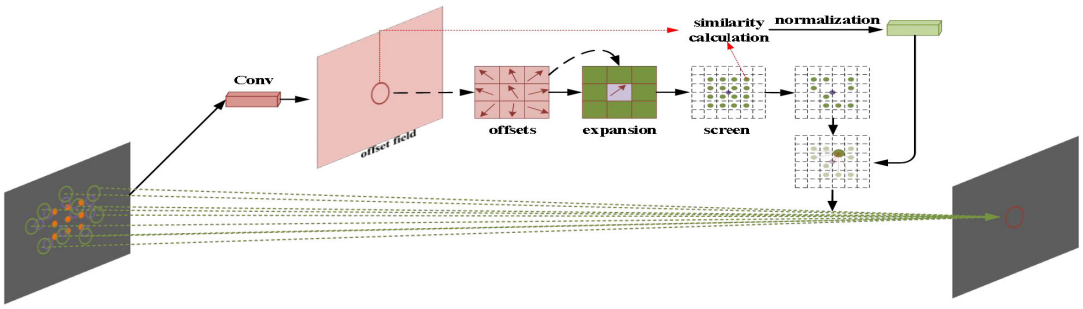


Fig. 5. Illustration of the 3 × 3 DDConv. Building upon deformation convolution, it extends to further diffuse secondary sampling points at distant locations, obtaining long-range dependencies by computing the similarity between secondary points and the central pixel.

remote sensing image processing, as illustrated in Fig. 5, we propose DDConv. This method improves upon DConv to simultaneously capture long-range dependencies while simulating variations in target morphology. Traditional convolution methods involve two steps: sampling on the input feature map x using a conventional grid $\mathcal{R}$ and assigning weights w to the sampling regions. Here, $\mathcal{R}$ defines a convolution kernel with a receptive field size of 3 × 3 and a dilation rate of 1

$$\mathcal{R} = \{(-1,-1),(-1,0),\ldots,(0,1),(1,1)\}.$$

Afterward, a sliding window of the same size is used to perform weighted summation over each pixel position p_0 in the input feature map x, resulting in the output feature map $\acute{x}$, where p_n represents the corresponding position of the input region $\mathcal{R}$ in the feature map, $\alpha(\cdot)$ denotes the weighted function of the self-learned weights

$$\acute{x}(p_0) = \sum_{p_n \in \mathcal{R}} \alpha(p_n) \cdot x(p_0 + p_n).$$

In the DDConv, a convolutional layer learns the offsets of the diffusion DConv kernel at the sampling points in the input feature map, namely the offsets Δp_n the vertical and horizontal coordinates of the grid $\mathcal{R}$. The sampling points with added offsets are defined as first-order sampling points p_n^1

$$p_n^1 = p_0 + p_n + \Delta p_n.$$

The sampling points with added offsets exhibit irregularity, and their pixel values are achieved through bilinear interpolation. The pixel value at position p_n^1 is calculated as follows:

$$V_{p_n^1} = \sum_q G\left(q, p_n^1\right) \cdot x(q)$$

where q represents the selection from all pixel positions in the feature map x, and $G(\cdot)$ denotes the bilinear interpolation function. To capture long-range target deformation dependencies, starting from the first-order point p_n^1 as the center, we diffuse around its spatial position to obtain second-order points p_n^2, forming the set U of second-order points

$$U = \text{Grid}_{4\times4}\left(\text{floor}\left(p_n^1\right) - 1\right).$$

Here, floor$(\cdot)$ denotes taking the largest integer not greater than the current value, $\text{Grid}_{4\times4}(\cdot)$ represents the pixel positions in the sampling region of the input feature map x for a 4 × 4 grid with the current point as the top-left corner, and the pixel values of the second-order points are denoted as $V_{p_n^2}$. The 16 obtained second-order points are compared with the current central position by calculating the cosine similarity based on pixel values. These points are then filtered using a threshold $\theta(\cdot)$, removing second-order points with lower similarity

$$\text{sim} = \theta\left(\text{Cosine}\left(V_{p_n^2}, V_{P_0}\right)\right).$$

The similarities are normalized to obtain the weight coefficients representing the relationships between the second-order and central points. These weight coefficients are then used to perform a weighted sum of the second-order points to derive the second-order aggregated information

$$V_{\boldsymbol{p}^2} = \sum_{\boldsymbol{p}_n^2 \in u} \text{SoftMax}(\text{sim}) V_{\boldsymbol{p}_n^2}.$$

Using the SoftMax$(\cdot)$ function, the second-order information is normalized. Finally, the second-order aggregated information is fused with the first-order aggregated information through addition to yield the final sampled value

$$V = V_{\mathbf{p}^2} + V_{\mathbf{p}^1}.$$

Clearly, analyzing and establishing connections for the feature information at each stage allows for capturing both the local deformation characteristics of the first-order offset points and the long-range dependencies of the second-order points. This approach also extends and enhances the deformation characteristics of the target objects.

C. Bidirectional Feature Distillation and Information Complementary Fusion Network

After obtaining the outputs of different modalities from the intermediate stages 2–4 of the backbone network, BDFusion is used to correspondingly fuse the features of the two modalities at different stages. This fusion is divided into two parts: one part relies on the bidirectional knowledge distillation interaction component to facilitate information exchange between the branches of the two modalities, employing a feature distillation strategy to enhance the interaction capability. The other part uses the adaptive spatial domain reconstruction component for adaptive spatial domain fusion of the features from different modality branches. Fusing features from different modalities is crucial for multimodal object detection tasks, and this is typically achieved by treating multiple modalities as equally

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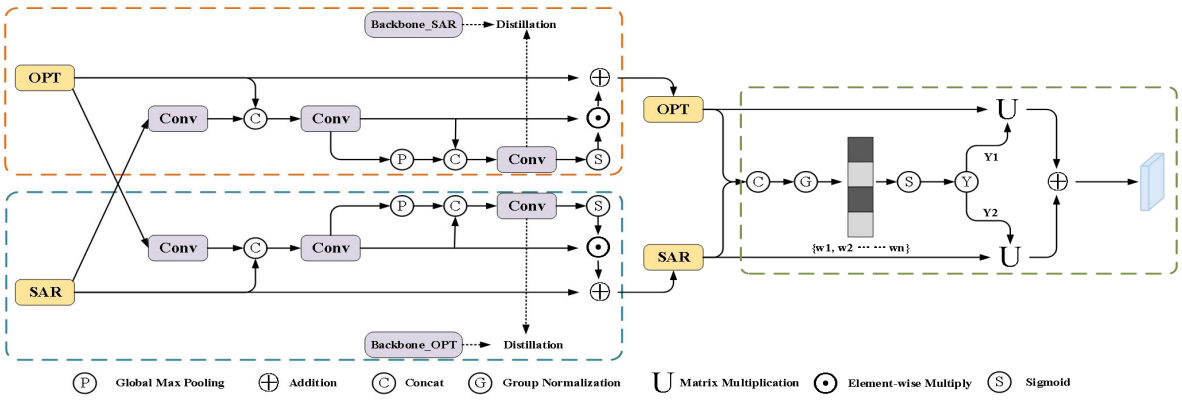


Fig. 6. BDFusion component structure diagram consists of two parts: bidirectional knowledge distillation interaction and adaptive spatial domain reconstruction module. The bidirectional knowledge distillation interaction (orange and blue line boxes) injects information from two different modalities into each other, facilitating information exchange between modalities and explicitly enhancing cross-modal utilization.

significant during the fusion process. However, due to the significant differences in the richness of information carried by different modalities, these balanced fusion methods cannot clearly distinguish the similarities and differences between features from different modalities. To explicitly differentiate and fuse features from different modalities, we propose the BDFusion. Fig. 6 illustrates the structure of this module, which mainly consists of two components: 1) bidirectional knowledge distillation interaction and 2) adaptive spatial domain reconstruction module. We will detail the implementation of these two modules in Sections III-C1 and III-C2.

1) Bidirectional Knowledge Distillation Interaction: The bidirectional knowledge interaction component, as shown on the left side of Fig. 6, is explained using optical and SAR imaging modalities with significant differences in their characteristics. This interaction comprises fusion blocks from optical (SAR) to SAR (optical). The optical branch learns features from SAR, and the SAR branch learns features from optical, enabling both modalities to fully understand the feature representation capabilities of the other modality, thereby explicitly enhancing the features of optical and SAR. The orange and blue boxes in Fig. 6 show that the fusion network structures from optical (SAR) to SAR (optical) are identical, with the only difference being the input positions. For simplicity, we provide a detailed description of the network structure using SAR feature injection into the optical branch (orange dashed line). After extracting contextual information through dilated convolution from the features of optical and SAR images output by the backbone, the SAR learning branch first acquires optical feature information

$$z = \text{Conv}_2(\text{Concat}(\text{Conv}_1(z_s), z_o))$$

where $\text{Conv}_1(\cdot)$ and $\text{Conv}_2(\cdot)$ represent standard convolutional layers, each consisting of a 1×1 convolutional layer, an activation function, and a regularization layer. Conv_1 is responsible for linearly transforming the SAR features, while Concat concatenates the two features along the channel dimension. Conv_2 aligns the dimensions of the concatenated features with the previous dimension of z_s, thus transforming the features into the residual space of SAR. Subsequently, we perform

$$w = \text{Gate}(\text{Conv}_3(\text{Concat}(\text{GMP}(z), z)))$$

where $\text{GMP}(\cdot)$ applies global max-pooling along the spatial dimensions to highlight important information in z. The information obtained from max-pooling is then concatenated back along the spatial dimensions through Concat operation. Conv_3 represents a standard convolutional layer responsible for aligning the dimensions of the concatenated features once again. Finally, using the Sigmoid activation function represented by $\text{Gate}(\cdot)$, attention weight coefficients ranging from 0 to 1 are generated. These weight coefficients are then used to weight and sum the SAR features z, extracting useful feature information and injecting it into the optical feature information through addition

$$\tilde{z}_o = (z \cdot w) + z_o.$$

To endow the knowledge injection branches of both modalities with stronger capabilities for extracting their modality features, an unsupervised feature distillation method is adopted, inspired by techniques such as knowledge distillation. Feature distillation and traditional knowledge distillation transfer knowledge from teacher to student models using final probability distributions. The selection of the teacher model and the design of the pretraining scheme are crucial for maximizing the effectiveness of feature distillation. To this end, considering the substantial differences in feature information among various modalities of remote sensing images, employing the same structured teacher model for knowledge transfer to the student model could lead to common features being extracted, which facilitates integration and fusion across different modalities. Conversely, the complex backgrounds and similar textures inherent in remote sensing images render their features relatively "fragile," whereby even minor changes in the network architecture can significantly affect the extracted features. Therefore, to ensure coherence in feature extraction, we utilize the dual-stream backbone network RepViIT [45] as the teacher model for each modality, while the student model selects the final convolution layer of the knowledge injection branch within the bidirectional knowledge distillation

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interaction framework. Additionally, to maintain model continuity, we first train DKDNet on the constructed multimodal dataset without employing the distillation strategy, using the respective modal feature extraction networks as pretrained teacher models. Regarding the selection of feature matching during the distillation process (the intermediate output features of both the teacher and student models), we choose the third stage of the backbone network as the intermediate output $F_t^3 \in \mathbb{R}^{192\times H/16} \times W/16$, taking into account the feature extraction capabilities of the teacher model and the receptiveness of the student branches

$$\text{Loss}_{\text{mse}} = L2(B_{\text{st}}(F_{\text{s}}) \cdot F_{\text{t}}).$$

F_{s} and F_{t} represent student features and teacher features, respectively, which are matched for distillation. The bridging module $B_{\text{st}}(\cdot)$ consists of a convolutional layer and batchnorm layer, used to match student features with teacher features. Finally, the mean squared error loss function with $L2$ distance measurement guides the student model to mimic the teacher model.

The total loss of the DKDNet model includes the regression and classification losses for the object detection task, as well as the optical modality feature distillation loss and the SAR modality feature distillation loss

$$\text{Loss} = w_1\text{Loss}_{\text{cls+bbox}} + w_2\text{Loss}_{\text{sar}} + w_3\text{Loss}_{\text{opt}}.$$

Among them, $\{w_1, w_2, w_3\}$ represents the weight ratios of different loss functions, and their impact on the model training performance will be discussed and analyzed in detail in the subsequent experimental section.

2) Adaptive Spatial Domain Reconstruction Component: To explore the implicit relationships between features from different modalities and better fuse the optical and SAR features after bidirectional knowledge interaction, we designed an adaptive spatial reconstruction component, as illustrated by the green dashed box in Fig. 6. Incorporating a gating selection mechanism to leverage the complementary information between different modalities, it adaptively selects spatial features with the highest information content to enhance detection and recognition accuracy. Specifically, given features from two different modalities optical $\tilde{z}_o, \tilde{z}_s \in \mathbb{R}^{C\times H\times W}$, we first concatenate the features along the channel dimension, then normalize the result by subtracting its mean and dividing by its standard deviation, as follows:

$$b = \text{Concat}\left(\tilde{z}_{\text{s}}, \tilde{z}_{\text{o}}\right)$$
$$\tilde{b} = \text{GN}(b) = \frac{b-\mu}{\sqrt{\partial^2+\varepsilon}}$$

where μ and ∂^2 represent the mean and variance, respectively, computed from the fused feature b, and ε denotes the offset. To provide a more flexible representation of weight coefficients, reflecting the diversity of pixel space, we initialize a trainable coefficient vector β. This vector is pointwise multiplied with the standardized features to obtain the final spatial reconstruction selection weights, indicating the importance of different modal feature maps in the spatial dimension

$$y = \text{Gate}\left(\text{Sigmoid}\left(\beta \cdot \tilde{b}\right)\right).$$

By applying the Sigmoid normalization function, the weight coefficients are mapped to the (0, 1) range. With a gate unit threshold set at 0.5, weight coefficients above the threshold are set to 1, indicating that the information in that region is acquired, while those below the threshold are set to 0, indicating that the information in that region is not acquired. The negation of the weight coefficients yields two opposing coefficients, y_{o} and y_{s}, which are, respectively, regarded as the selection coefficients for optical and SAR features. Finally, the input optical and SAR features $\tilde{z}_{\text{o}}$ and $\tilde{z}_{\text{s}}$ are multiplied by y_{o} and y_{s} coefficients, respectively, and then added together to obtain the final multimodal adaptive spatial reconstruction fusion feature

$$f = (\tilde{z}_{\text{s}} \cdot y_{\text{s}}) + (\tilde{z}_{\text{o}} \cdot y_{\text{o}}).$$

Pixels in spaces exceeding this threshold acquire optical features, while those below acquire SAR features.

IV. Experiments

A. Data Description and Evaluation Metrics

In this section, extensive experiments are conducted on the RGB + IR dataset VEDAI [21] and our proposed RGB + SAR multimodal OSSDD dataset to validate the practical performance of our proposed DKDNet multimodal object detection algorithm in real-world scenarios.

1) VEDAI: VEDAI [21] is a publicly available dataset for vehicle detection in multimodal remote sensing images. Each image is captured from aerial views and contains two modalities: RGB and IR. The dataset comprises over 1200 images containing over 3700 annotated object instances. The size distribution of object bounding boxes is illustrated in Fig. 7(a). The resolution of each image varies between 1024 × 1024 and 512 × 512 pixels. The dataset includes diverse backgrounds such as forests, parking lots, mountains, urban roads, farmland, and nine vehicle categories (car, camper, pickup, tractor, other, truck, ship, van, and plane). The distribution of instances for each category is depicted in Fig. 7(b). The objects in the images are small vehicles, each vehicle occupying approximately 0.2% of the total image pixels on average and having a spatial resolution ranging from 10 to 30 cm. The original annotations, initially in the format of rotated bounding boxes represented by four corner coordinates, were converted to horizontal bounding box format. Following the validation method outlined in VEDAI [21], the tenfold cross-validation protocol was adopted. In our experiments, we choose images of size 1024 × 1024 pixels for both training and testing. We ignore the detection of the "Plane" and "Ship" classes and only focus on detecting various vehicle categories of interest.

2) OSSDD: Due to the absence of publicly available object detection datasets with optical and SAR registration, we constructed an optical and SAR corresponding ship slice dataset from satellite images of the Suez Canal region to investigate the fusion detection performance of our method across these two significantly different modalities. The optical images in OSSDD are acquired from the European Space Agency's Sentinel-2 multispectral satellite, with a spatial resolution of approximately 10 m. SAR images are obtained from the

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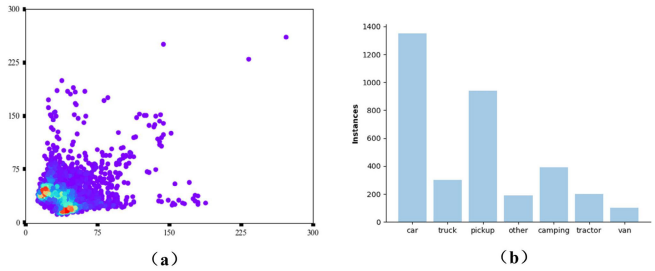


Fig. 7. (a) Distribution of object bounding box sizes (Pixel) and (b) instance counts for various classes.

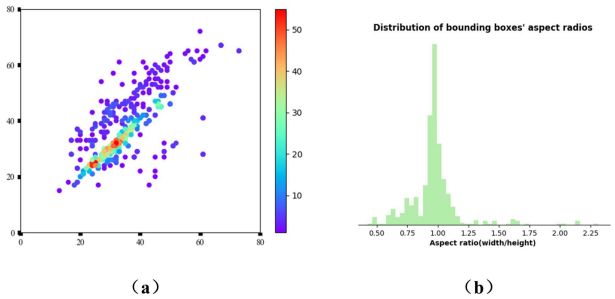


Fig. 8. (a) Distribution of object bounding box sizes (Pixel) and (b) aspect ratios.

Sentinel-1 satellite, with a spatial resolution ranging from 5 to 10 m. We registered the optical and SAR images using professional registration tools to generate three large-scale registered images. Domain experts manually annotated the ground truth of ships based on the optical images and prior information. We sliced the images into overlapping regions, resulting in over 1000 pairs of 256 × 256 optical and SAR ship slices, containing more than 700 ship targets. We randomly divided the dataset in a 5:1 ratio, resulting in a training set comprising 1027 pairs of images and a test set containing 205 pairs of images. The size and aspect ratios of each ship target's bounding box in pixels are illustrated in Fig. 8. The ships in OSSDD include both nearshore and offshore targets and are affected by natural conditions such as cloud and fog, posing significant challenges for multimodal detection tasks. To further investigate the robustness of our proposed method compared to other deep learning approaches under a significantly reduced number of training samples, we will reduce the training set size by half for training and testing.

3) Evaluation Metrics: This article quantitatively evaluates and compares the performance of different detection methods on multimodal object detection datasets using various mainstream evaluation metrics for detection and recognition tasks. These metrics include Precision, Recall, mAP0.5, and mAP0.95. These detection standards are obtained through four classification problem metrics: true positives (TPs), false positives (FPs), true negatives (TNs), and false negatives (FNs). The calculation formulas are as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$
$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

AP is the ratio of the area under the precision-recall curve to the total area. Higher AP indicates that the algorithm detects more target instances accurately. The calculation formula is as follows:

$$\text{AP}_{\text{IOU}} = \int_0^1 \text{Precisiond(Recall)} = \int_0^1 p(r)dr$$
$$\text{mAP} = \frac{\text{AP}}{N} = \frac{\int_0^1 p(r)dr}{N}.$$

Intersection over union (IOU) represents the threshold of the intersection over the union ratio between the predicted box and the ground truth box. If this ratio exceeds the threshold, the predicted box is considered a positive sample. $\text{AP}_{0.5}$ refers to the AP at an IoU threshold of 0.50. mAP0.5 denotes the average of all AP values for all classes at an IoU threshold of 0.50. $\text{mAP}_{0.5:0.95}$ is the primary challenging metric, which means the mAP at IoU thresholds ranging from 0.50 to 0.95. For convenience, it is commonly abbreviated as mAP and is much stricter than $\text{AP}_{0.5}$.

B. Implementation Details

Our experiments were conducted on a GPU workstation equipped with Ubuntu 21.0, CUDA (10.0), and cuDNN 7.0, featuring an NVIDIA RTX 3090 with 24 GB of memory. The software environment included PyCharm 2021, PyTorch 1.11, and Python 3.8. Several data augmentation techniques, such as affine transformations of images, mainly including rotation, scaling, translation, and cropping, were implemented during the training process. Mosaic data augmentation and other techniques were also employed.

During the training phase, we set the number of epochs to 300 and the batch size to 8. We utilized stochastic gradient descent (SGD) for network optimization, with an initial learning rate of $1e-2$, a momentum of 0.937, and a weight decay $5e-4$. The input image size was set to 640 × 640. Our DKDNet detection network employed YOLOv8s as the base detector. Neither our model nor the compared algorithms utilized pretrained models in the backbone section.

C. Ablation Study

1) VEDAI: Quantitatively, both Tables I and II demonstrate the detection performance of our method compared to several competitive methods on the VEDAI [21] dataset. Visually, as shown in Fig. 9(a), our method consistently outperforms the other four methods in validation accuracy across different training iterations, exhibiting more stable precision curves. Our DKDNet detection method surpasses other networks, as indicated in Table II, achieving a mAP0.5 score of 77.9%, which is 1.7% higher than the next best method, SuperYOLO [25], and significantly higher than the baseline model YOLOv8's 73.7%. Similarly, as shown in Table I, our method also achieves a mAP of 49.2%, over 2% higher than the baseline model. Given that prior research by Liu et al. [9] has validated the effectiveness of the dual-branch structure for extracting features from multimodal images, we adopt a dual-branch backbone network for multimodal detection using YOLOv5s and YOLOv8s. Compared to YOLOv5s and YOLOv8s methods designed

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TABLE I

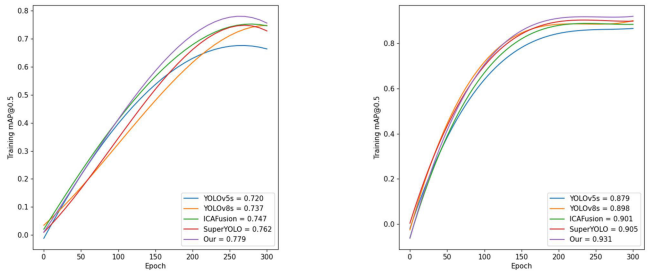
COMPARISONS OF PERFORMANCES WITH DIFFERENT DATASETS IN TERMS OF PRECISION, RECALL, MAP0.5, AND MAP. ALL SCORES ARE EXPRESSED AS PERCENTAGES (%)

Dataset	Modality	Backbone	method	Precision ↑	Recall ↑	$mAP_{0.5}$ ↑	mAP ↑
VEDAI	Multi	CSPDarknet53	YOLOV8s	78.8	65.4	73.7	46.1
	Multi	RepViT[45]	YOLOV8s	80.1	67.0	74.8	47.2
	Multi	CSPDarknet53	DKDNet	80.6	68.3	75.3	47.7
	Multi	RepViT[45]	DKDNet	**82.4**	**70.6**	**77.9**	**49.2**
OSSDD	Multi	CSPDarknet53	YOLOV8s	92.4	80.7	89.8	51.8
	Multi	RepViT[45]	YOLOV8s	93.3	82.9	90.3	52.6
	Multi	CSPDarknet53	DKDNet	95.0	85.3	92.0	55.4
	Multi	RepViT[45]	DKDNet	**97.2**	**86.7**	**93.1**	**57.0**

TABLE II

QUANTITATIVE ANALYSIS OF VEDAI TEST SET WITH STATE-OF-THE-ART MODEL RESULTS, PERFORMANCE IS EVALUATED IN TERMS OF MAP0.5

Method	Modality	Car	Pickup	Camping	Truck	Other	Tractor	Van	$mAP_{0.5}$ ↑
YOLOv5s	RGB	67.3	54.6	61.8	71.1	44.9	70.5	75.3	66.7
YOLOv5s	IR	64.6	51.9	52.6	69.7	42.6	60.1	70.5	63.2
YOLOv5s	Multi	80.9	71.6	67.3	82.1	52.0	78.6	80.2	72.0
YOLOv8s	RGB	78.5	70.6	64.9	78.5	50.7	76.0	75.4	69.8
YOLOv8s	IR	71.2	54.8	57.3	63.8	43.0	62.9	68.5	65.1
YOLOv8s	Multi	82.7	73.6	69.8	82.4	53.3	80.0	81.5	73.7
ICAFusion[15]	Multi	82.3	74.9	70.8	84.6	57.6	82.4	83.0	74.7
SuperYOLO[25]	Multi	87.9	76.0	72.0	85.5	55.8	82.0	82.9	76.2
Our	Multi	**88.1**	**76.3**	**77.2**	**87.8**	**60.4**	**83.0**	**84.7**	**77.9**



(a) (b)

Fig. 9. $mAP_{0.5}$ precision smooth curves of DKDNet and other comparative methods on two datasets with different training iterations. (a) VEDAI. (b) OSSDD.

based on single-source images, our use of the BDFusion component effectively injects features from different modalities, facilitating information flow between modalities. Comparison with state-of-the-art multimodal detection methods such as ICAFusion [15] and SuperYOLO [25], our approach, by considering the flow of information between modalities and the benefits of long-term interaction, maintains a leading position after gaining these benefits. For objects with small shape variations and low discriminability, such as cars and pickups, detection accuracy reaches 88.1% and 76.3%, respectively. This is attributed to the sensitivity of our proposed DDConv to subtle geometric shape differences in targets, enhancing interclass discrimination. Simultaneously, through diffusion mechanisms, distant dependencies are captured, enhancing feature representations of similar objects.

As shown in rows 2–7 of Table II, we conducted three sets of experiments on YOLOv5 and YOLOv8 to further explore the necessity of multimodal fusion. Results indicate that when detecting using only single-source data such as IR, YOLOv5, and YOLOv8 achieve mAP0.5 scores of only 63.2% and 65.1%, respectively. Even under favorable lighting conditions, although optical geometric features are rich, using only RGB images for detection yields accuracies of only 66.7% and 69.8%, respectively. Compared to single-source methods, even without designing any multimodal fusion modules, simply fusing multimodal features through a straightforward channel stacking approach increases the mAP0.5 metric by at least 5.3% and 3.9%, respectively, compared to using only RGB or IR images.

To present a more intuitive comparison between our method and others regarding detection performance, Fig. 10 illustrates the detection results for different vehicle categories in various scenarios. Several representative samples were chosen for visual comparison. Among them, samples marked as Fig. 10(a), (b), (h), and (i) represent multicategory vehicles on complex urban roads; Fig. 10(f) and (g) depicts vehicles in large parking lots; while Fig. 10(c)–(e) depicts vehicles traveling on wide roads. From Fig. 10, it can be observed that our DKDNet can detect almost all vehicles, while other methods tend to exhibit varying degrees of false positives or misses.

2) OSSDD: The experiments on the OSSDD dataset and the VEDAI [21] dataset validate the effectiveness of our method in RGB and IR image fusion detection. Tables I and III present the quantitative experimental results of our method compared to several competitive methods on the OSSDD dataset. Visually, as observed in Fig. 9(b), our method consistently outperforms the other four methods in validation accuracy when the training count exceeds 150. Table III shows that our method achieves a high mAP0.5 score of 93.1%, surpassing the second-ranked method SuperYOLO [25] by 2.6% and significantly outperforming the baseline model YOLOv8 at 89.8%. Comparing the single-modal and multimodal contrasts between YOLOv5 and YOLOv8, it can be seen that using SAR images alone for detection is far less

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Fig. 10. Qualitative comparison of RGB + IR object detection on the VEDAI test set with other methods, with detected objects indicated by arrows. (a)–(i) Nine representative samples of different scenes for visual comparison.

TABLE III

QUANTITATIVE ANALYSIS OF OSSDD TEST SET WITH STATE-OF-THE-ART MODEL RESULTS, PERFORMANCE IS EVALUATED IN TERMS OF GFLOPS PARAM AND MAP0.5

Method	Modality	GFLOPs ↓	Param ↓	$mAP_{0.5}$ ↑	$mAP_{0.5}$(small) ↑
YOLOv5s	RGB	**13.7**	7.0	85.4	82.0
YOLOv5s	SAR	**13.7**	7.0	79.0	74.3
YOLOv5s	Multi	20.4	14.3	87.9	84.2
YOLOv8s	RGB	36.6	11.1	86.5	82.7
YOLOv8s	SAR	36.6	11.1	81.3	75.8
YOLOv8s	Multi	54.4	18.0	89.8	86.7
ICAFusion[15]	Multi	34.5	120.2	90.1	88.2
SuperYOLO[25]	Multi	18.5	**5.1**	90.5	88.4
Our	Multi	70.4	19.2	**93.1**	**91.5**

effective than using RGB and multimodal detection due to the lack of geometric, textural, and color information in SAR images.

We have added two evaluation metrics (Param and GFLOP) in Table III. The table shows that our model's parameter count is only slightly higher than that of YOLOv5, YOLOv8, and SuperYOLO [25]. This increase can primarily be attributed to selecting RepViT [45] as the backbone network, which inherently contributes more parameters to the overall model. Additionally, the proposed DDConv necessitates the computation of offsets for each convolutional position, and to capture long-range interaction information, we have expanded the convolutional points to include second-order sampling. This extension results in an increased computational burden to adequately extract long-distance information, thereby yielding a higher GFLOPs metric compared to other methods. However, the trade-off of a relatively modest increase in computational load for a significant enhancement in accuracy is deemed acceptable, as our model achieves an optimal balance between speed and accuracy. As shown in Table I,

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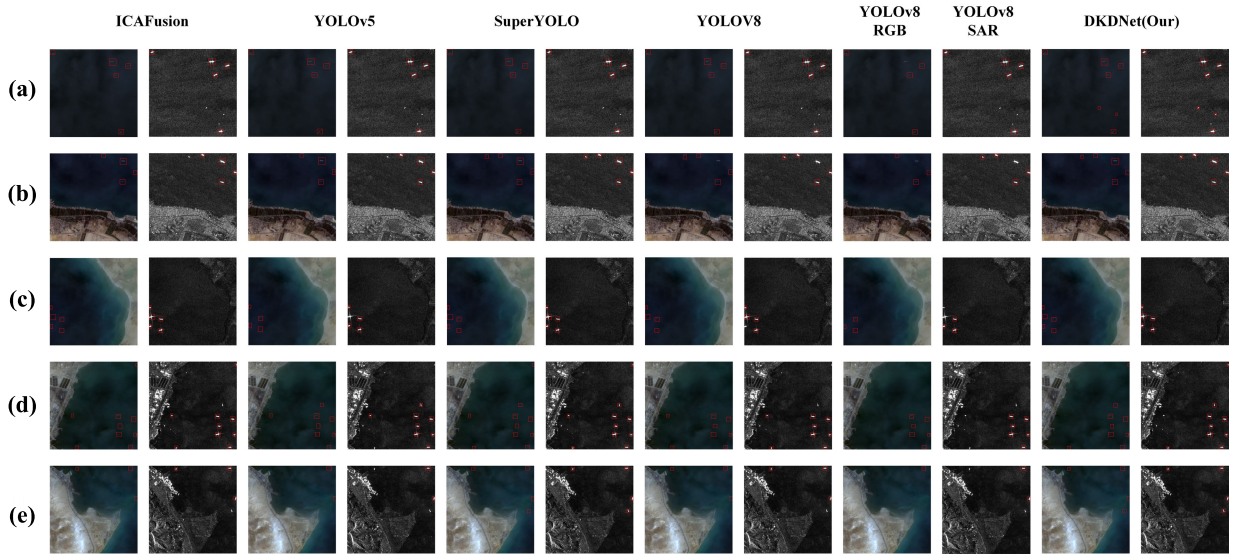


Fig. 11. Qualitative comparison of RGB + SAR object detection on the OSSDD test set with other methods. (a)–(e) Five representative samples of different scenes for visual comparison.

our method achieves a 57.0% mAP, which is 4.4% higher than the baseline model YOLOv8, even under a stricter mAP criterion.

To investigate the impact of training set size on the performance of our method, we randomly selected half of the images from the original training set for experimental evaluation of our model and other methods. As shown in Table III $mAP_{0.5}$ (small) column, our method achieves optimal accuracy even with limited training samples. The robustness of our method concerning the size of the training data can be attributed to two key factors. First, the designed DDConv facilitates a global understanding by effectively utilizing contextual information. Second, implementing knowledge distillation enhances the information injection between modal features, enabling the transfer of substantial knowledge learned by the teacher model to the knowledge injection branches, which aids in rapid learning from a limited dataset. Additionally, the proposed adaptive spatial component allows for the dynamic selection of different modal features for fusion, thereby uncovering implicit relationships among multimodal features in the spatial dimension and enhancing the model's generalization ability. Fig. 11 illustrates the detection performance of our DKDNet compared to other methods on the OSSDD dataset.

3) Impact of Knowledge Distillation Strategy: To validate the effectiveness of the knowledge distillation strategy in the BDFusion component, we conducted extensive experiments involving two datasets. As shown in Fig. 12, employing the knowledge distillation strategy resulted in an improvement of 2.5% and 3.1% in mAP metrics on the two datasets, respectively. In this strategy, we utilized the intermediate outputs of the pretrained backbone network as the teacher model. In contrast, the intermediate outputs of the modality injection branch served as the student model. By guiding the student model with the teacher model, we enhanced the current modality representation capability and injected it as supplementary information into the other branch.

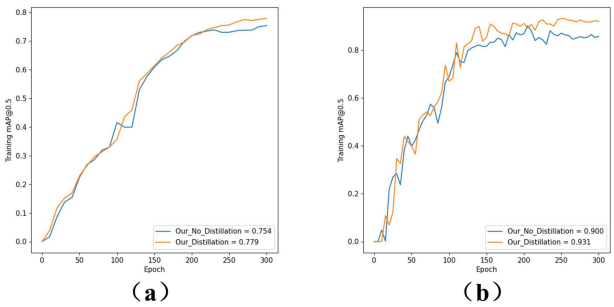


Fig. 12. Detection accuracy curves of DKDNet with and without knowledge distillation strategy on two datasets. (a) VEDAI. (b) OSSDD.

To further investigate the impact of knowledge distillation on the internal structure of the BDFusion component, as depicted in Fig. 13, we present the visualization results of the intermediate outputs of the BDFusion component on the OSSDD test set. It is evident from the second column that employing the knowledge distillation strategy enhanced the representation capability of both modal branches, resulting in higher brightness of the targets.

4) Impact of Different Convolution Methods: We evaluate our proposed DDConv against various other convolutional methods. As shown in Table IV, DDConv outperformed other convolutions on the VEDAI [21] dataset. From the table, it can be observed that using only traditional 3×3 convolutions resulted in the lowest detection accuracy, attributed to the fixed local receptive field of convolutions. To enlarge the receptive field, we replace conventional convolutions with dilated convolutions with a ratio of 6, resulting in a 0.3% improvement in mAP, effectively demonstrating the significance of long-range dependencies. When utilizing DConvs, the mAP metric reached 48.5%, a 1.5% improvement over regular convolutions, owing to the ability of DConvs to better learn object deformations, making them more adaptable to

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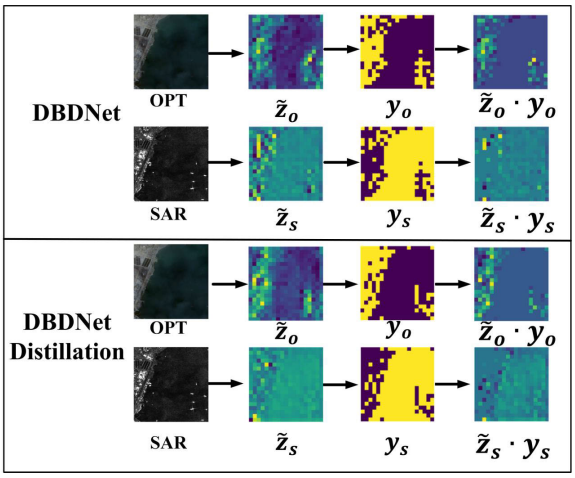


Fig. 13. Visualization of intermediate outputs of the internal structure of the BDFusion module on the OSSDD test set. The upper and lower parts of the figure represent BDFusion with and without knowledge distillation strategy, respectively. The first column shows the input images paired with optical and SAR. The second column depicts the optical and SAR feature maps after knowledge injection, with brighter colors indicating greater network attention to the current region. The third column displays the adaptive spatial coefficients for both modalities. Since we employ hard-threshold filtering, the selection coefficients for optical and SAR feature fusion are complementary. The fourth column shows the selected optical and SAR features after hard gating.

TABLE IV

COMPARISON OF DIFFERENT CONVOLUTION TYPES AND PARAMETER SETTINGS ON THE VEDAI

Type of Convolution	mAP ↑	$mAP_{0.5}$ ↑
$Convolution_{3\times3}$	47.0	74.5
Dilation $Convolution_{3\times3}$	47.3	75.4
Deformable $Convolution_{3\times3}$	48.5	77.2
$DDconv_{3\times3}$	**49.2**	**77.9**
$DDconv_{5\times5}$	47.5	73.7
$DDconv_{7\times7}$	47.3	72.0

TABLE V

ABLATION EXPERIMENTS ON THE VEDAI DATASET TO SEQUENTIALLY VALIDATE THE EFFICACY OF EACH COMPONENT

DDConv	BDFusion		Precision ↑	Recall ↑	$mAP_{0.5}$ ↑
	Two-way knowledge distillation interaction	Adaptive spatial domain reconstruction			
			80.1	67.0	74.8
✓			80.3	67.6	75.1
	✓		81.6	68.3	75.8
		✓	81.2	68.0	75.5
✓	✓		81.9	68.9	76.8
✓		✓	81.3	68.2	76.2
	✓	✓	82.1	69.7	77.0
✓	✓	✓	**82.4**	**70.6**	**77.9**

object variations. Our proposed DDConv exhibits sensitivity to various object geometries while capturing long-range dependencies, thus achieving optimal performance.

5) Effects of Different Components: This section discusses the impact of two components within DDConv and BDFusion—two-way knowledge distillation interaction and adaptive spatial domain reconstruction—on DKDNet. Table V presents the results of these two proposed components on the VEDAI [21] dataset, with experiments using YOLOv8 as the baseline backbone network, following RepViT [45].

TABLE VI

IMPACT OF THE COMBINED WEIGHTING COEFFICIENTS OF LOSS FUNCTIONS $\{w_1, w_2, w_3\}$ ON THE OSDD DATASET

w_1	w_2	w_3	$mAP_{0.5}$ ↑
1	0	0	91.2
0.5	0.5	0	91.9
0.5	0	0.5	92.3
0.3	0.3	0.3	92.0
0.2	0.4	0.4	91.7
0.2	0.6	0.4	91.4
0.2	0.2	0.6	91.6
0.8	0.1	0.1	**93.1**
0.5	0.25	0.25	92.9

The results indicate that using only DDConv resulted in a modest performance improvement of just 0.3%. However, utilizing only either the two-way knowledge distillation interaction or the adaptive spatial domain reconstruction component within BDFusion led to performance gains of over 0.7%, while combining both components yielded an improvement of 2.2%. With both two-way knowledge distillation interaction and adaptive spatial domain reconstruction components working together, multimodal information can be fully utilized, generating fusion feature maps with greater information content. Additionally, combination experiments were conducted with both DDConv and BDFusion components, as shown in rows 5 and 6 of the table. Following DDConv, multilayer feature maps captured various deformation features and long-range dependencies of targets, thereby enhancing their representation capability. Compared to using only the two-way knowledge distillation interaction or the adaptive spatial domain reconstruction component separately, performance improved by 1.0% and 0.7%, respectively. As depicted in Fig. 14, we showcase visualization examples of attention maps for intermediate outputs of each component within DKDNet on the VEDAI [21] test set.

6) Selection of Combination Weight Hyperparameters $\{w_1, w_2, w_3\}$ for the Loss Function: The weighting coefficients $\{w_1, w_2, w_3\}$ of the losses are critical for balancing the object detection performance and the multimodal feature learning in the knowledge injection branch. As shown in Table VI, we trained and tested with different combinations of values on the OSSDD dataset. From the first three rows of the table, it can be observed that regardless of which branch is enhanced by knowledge distillation, the detection accuracy improves by at least 0.7%, which demonstrates the effectiveness of the knowledge distillation strategy in strengthening the feature learning of multimodal branches, thereby enriching the subsequent fused features. To explore the optimal combination of weight coefficients, the last six rows of the table show the results of experiments with various weight combinations. It is evident that the network achieves the best performance when the weights $\{w_1, w_2, w_3\}$ are set to 0.8, 0.1, and 0.1, yielding a mAP0.5 of 93.1%. From the final row, it can be seen that reducing this ratio to 0.5, 0.25, and 0.25 results in a 0.2% drop in performance. When the loss ratios are equal across all three components, the performance reaches 92.0%. Therefore, we selected the $\{w_1, w_2, w_3\}$ values of 0.8, 0.1, and 0.1, and these were kept constant in the experiments.

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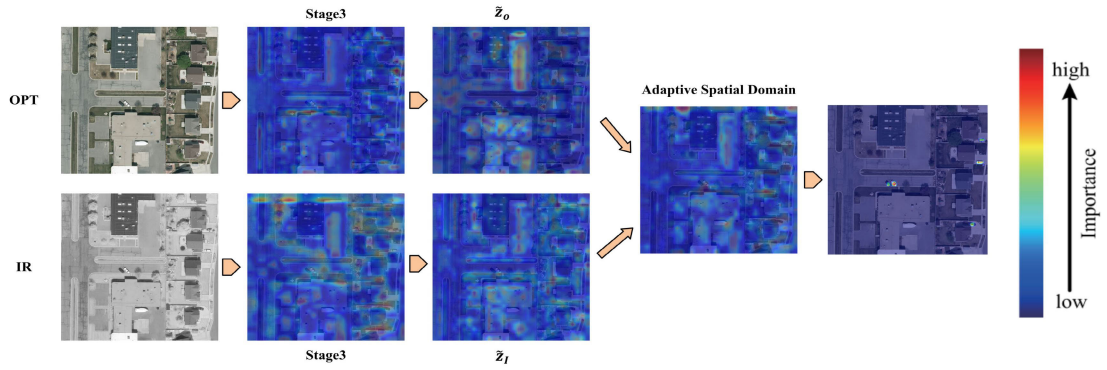


Fig. 14. Presents an example of visualized attention maps for intermediate outputs of various components in DKDNet on the VEDAI test set.

V. Conclusion

In this article, to leverage complementary information between different modalities, we propose a DKDNet for multimodal object detection tasks. Specifically, we use RepViT as the backbone network for dual-stream feature extraction of different modalities. Additionally, we introduce a bidirectional feature distillation information fusion network, which injects information from another modality into the current modality via bidirectional knowledge injection branches to facilitate information flow between modalities. Subsequently, we utilize an adaptive spatial domain reconstruction module to generate spatial weighting coefficients for different modal features. Experiments are conducted on our constructed optical and SAR multimodal ship target detection dataset, as well as the publicly available RGB + IR vehicle detection dataset VEDAI. Our DKDNet achieves state-of-the-art performance on the optical dataset compared to other advanced methods. In future research, we will further explore lighter and more efficient object detection networks. Meanwhile, we will extend multimodal detection methods to single-modal applications to meet the demands of real-time detection tasks in practical scenarios.

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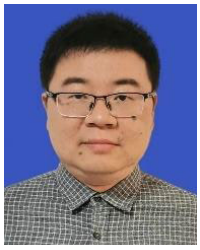
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